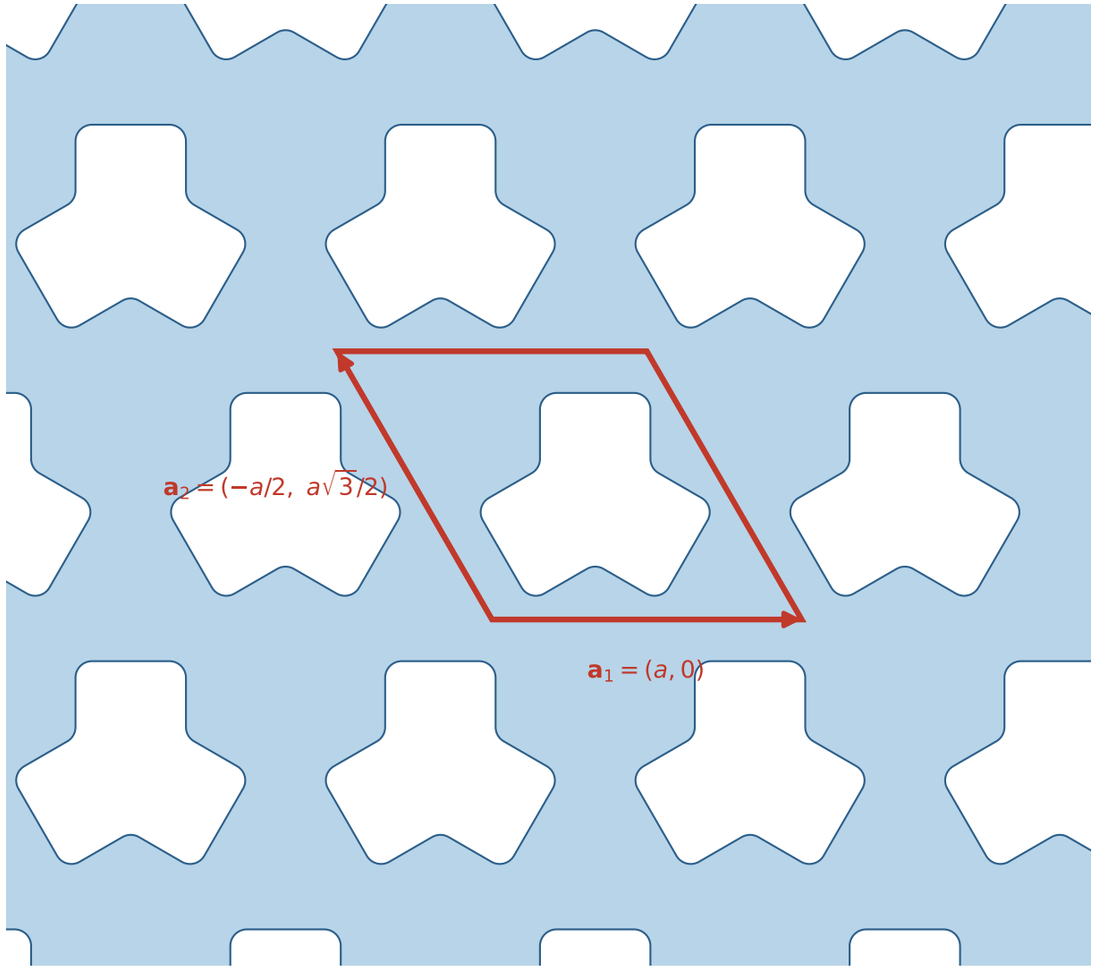
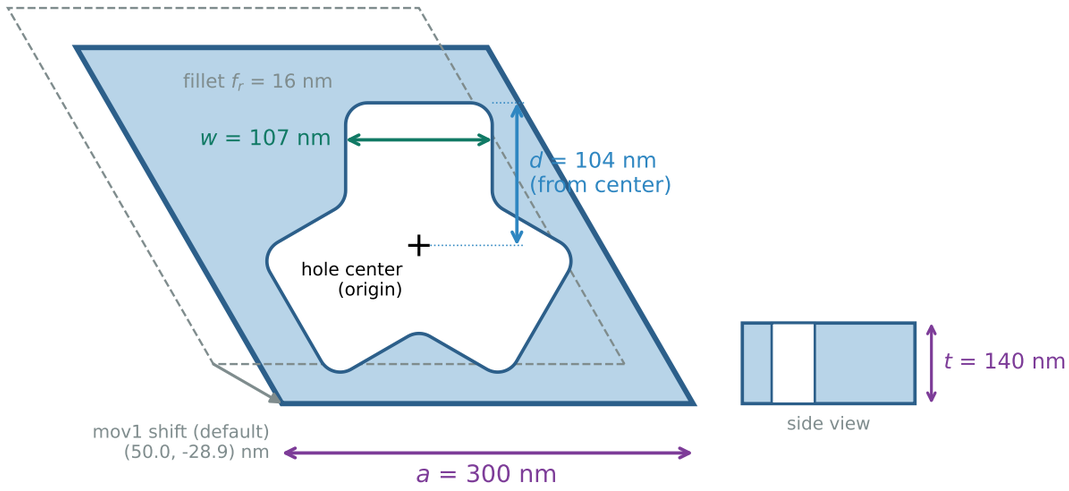
# Diamond snowflake phononic crystal -- diamond_mech_v1 simulation setup
**unit_cell_3.mph | a=300, d=104, w=107, f_r=16, t=140 nm | E=1050 GPa, nu=0.07, rho=3500 kg/m^3 | N_mode=20, f_mech=10 GHz**

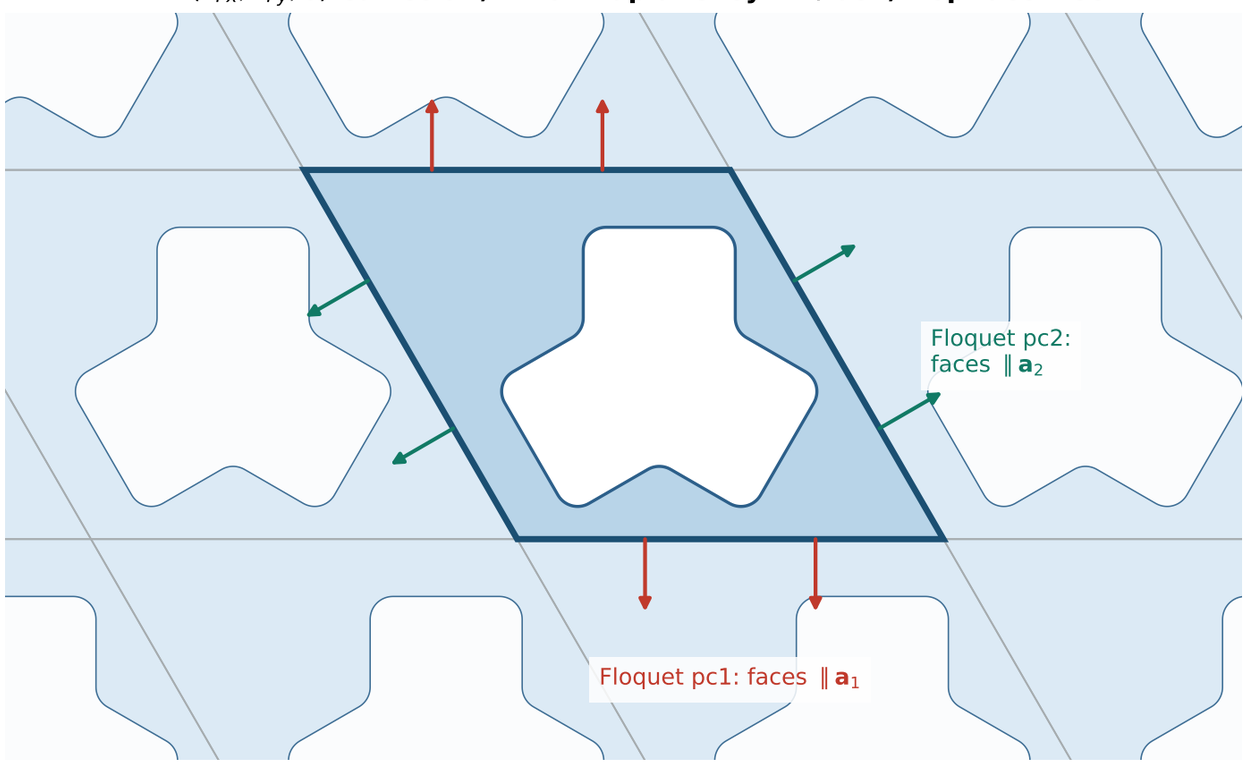
## (A) Triangular lattice -- rhombus primitive cell (as in pol1)



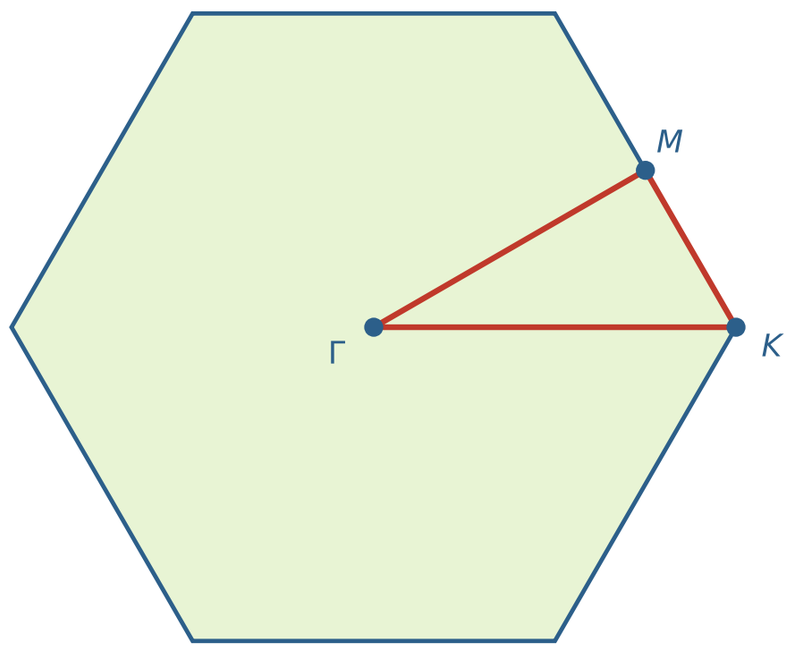

## (B) Unit cell (top view): hole at lower-triangle centroid, 3 arms of length d from the center at 90/210/330 deg




## (C) COMSOL cell + Floquet BCs (pc1, pc2)
**$\mathbf{k} = (k_{Fx}, k_{Fy}, 0)$ Cartesian; z=0 midplane: sym1 / as1; top face free**




## (D) First Brillouin zone + k-path




$K = (4\pi/3a,\ 0)$ vertex; $M = (\pi/a,\ \pi/\sqrt{3}a)$ edge midpoint
band path $\Gamma \to M \to K \to \Gamma$

Geometry drawn from unit_cell_3.mph feature definitions (r1, rot1/2, fil1, pol1, mov1 in wp1+wp2), read via mph on 2026-10-14.